Algebra 1
Unit 7
Lesson 7 Homework

Name____Answer key_____
Date____________________
Period__________________

For questions 1-2, consider the perfect square trinomial $9x^2 + 24x + 16$.

1. Rewrite the trinomial in the form $a^2 + 2ab + b^2$.

   $(3x)^2 + 2(3x)(4) + 4^2$

2. Factor the perfect square trinomial.

   $(3x + 4)^2$

For questions 3-4, consider the perfect square trinomial $16x^2 - 40x + 25$.

3. Rewrite the trinomial in the form $a^2 - 2ab + b^2$.

   $(4x)^2 - 2(4x)(5) + 5^2$

4. Factor the perfect square trinomial.

   $(4x - 5)^2$

For questions 5-8, determine whether the quadratic expression is a perfect square trinomial. Then, factor the expression.

5. $w^2 - 20w - 100$

No.
Cannot be factored

6. $f^2 - 2fg + g^2$

Yes.
$(f - g)^2$

7. $25v^2 - 9$

No.
$(5v + 3)(5v - 3)$

8. $9y^2 - 81y + 18$

No.
$9(y^2 - 9y + 2)$

For questions 9-10, factor the trinomial.

9. $25a^2 + 120a + 144$

$(5a + 12)^2$

10. $p^2 - 22p + 121$

$(p - 11)^2$